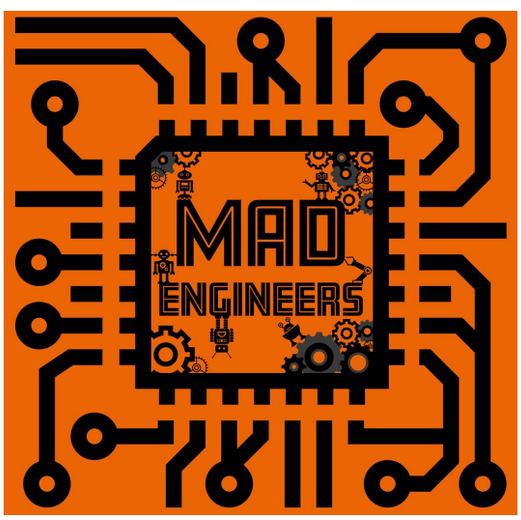
# MAD ENGINEERS




## OVER THE COUNTER CONTACTLESS SUPPLY PACKAGING AND DELIVERY, CURRENCY AND TABLE DISINFECTION ROBOT

| | |
|---|---|
| **LOCATION** | Mumbai, India |
| **SCHOOL** | On My Own Technology |
| **TEAM NO.** | FTC #2123 |

# Executive Summary

- *Our problem statement is simple*: **essential supplies workers,** such as pharmacists, supermarket cashiers and security guards, face a **higher risk** of contracting infectious diseases such as COVID-19. For example, in the United States, at least **41 grocery store workers** have died from the disease.
- *Our robot is designed with two simple objectives in mind*: a) to eliminate **every form of surface transmission of pathogens** b) to help enforcement of **physical distancing measures.**
- *Our robot achieves this by:* autonomously implementing a system of goods transfer that **prevents human-human and human-robot contact**, while simultaneously sanitising the **currency it transfers and the table it moves on.**

# Problem Statement

Let's consider the example of the COVID-19 pandemic. While governments and citizens have taken exemplary measures to help essential supplies workers, these efforts are falling short.

- In Spain, **276 pharmacy workers** had to take time off after "contracting or coming into contact with coronavirus."
- In Varanasi, India, a **single infected pharmacist caused 14 other cases**, amongst them two other pharmacists. This demonstrates the risk infected essential supplies workers can **pose to the many people they come in contact with on a daily basis.**
- Concerns have risen over the possibility of the virus **spreading through banknotes** - countries like the United States are **quarantining the cash coming in from Asia.**
- New York City and CDC health guidelines state that it is essential to disinfect "**high-touch surfaces like desks and countertops.**
- Furthermore, according to a technical paper in the BIS bulletin, "frequently-touched objects, such as **credit card terminals or PIN pads**" pose a greater risk. This is backed up by CDC health guidelines which recommend sanitisation of "**point of sale keypads"**.

We come from India. It's distressing enough to see our developing country **struggle** under the load of the virus. It's even more disturbing that the majority of the brunt is borne by **common working-class Indians**, like essential supplies workers.
India has a pharmaceutical worker density of just **0.5 per 1000 people**. About **95% of retail** happens through traditional brick and mortar stores, which has forced *local kirana stores* to take up a large part of Indian demand. These statistics alone spell a dangerous picture for India - and the **risk is even more if we don't take basic precautions** - physical distancing, table disinfection, card machine sanitisation, and preventing human-human contact.
Our over-the-counter robot has **four unique value propositions** that tackle this problem.

- **TABLE DISINFECTION**: Our robot **regularly disinfects** the table - so that if a customer unwittingly places their smartphone or hands on the table, they remain safe from the virus.
- **CONTACTLESS**: Our robot **prevents all forms of human-human contact and human-robot contact.** The supplies being transferred do not even touch the table - they are transported in a supplies box. Here, they are automagically **packaged in an elastic bag**, which the customer can pick up.
- **SAFE CURRENCY**: Our robot **eliminates cash, coins, bill and credit card machines** as a vector for the virus - safely sanitising every note and coin. We have a built-in credit card machine and billing machine. They keypad of the credit card machine is sanitised every single transaction.
- **PHYSICAL DISTANCING**: Our robot **enforces social distancing norms**. Due to the mobility of the robot, **workers can stand at a safe distance** while the robot transfers the supplies to the customer. It also **normalises separate distanced booths for customers.**

# About our Alliance

For the Robots to The Rescue competition, **MAD Engineers #2123, G-Force #2142, and R-Factor #6042** are coming together in an alliance.
Our alliance theme is **assisting frontline workers**. Team MAD Engineers tackles this problem on a **specialised, focused level,** assisting essential supplies workers in times of pandemic. Team R-Factor's pothole-filling mechanism helps **municipal cooperation workers** in ensuring roads remain smooth and efficient. Team G-Force attacks the problem on a **broad, global level** by helping frontline workers in all kinds of disaster response scenarios. In this way, both teams were able to solve the issue of global emergencies on different levels using different approaches.
All teams provided each other with "***help sessions***". **Team G-Force** provided us with OnShape tips and tricks, while we helped them with documentation and presentation. **Team R-Factor** helped us optimise our robot design and team collaboration workflow, while we suggested unique ideas for their robot, like the installation of pneumatic wheels.

# Design Process

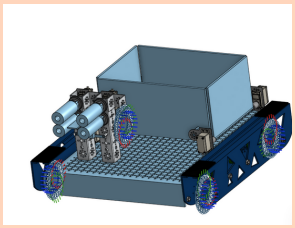
## STAGE 1: IDEATION AND DEBATE

Our greatest struggle was in striking a balance between our unabashed optimism and the sobering nature of reality. Our challenge was to create a minimal, robust, and inexpensive robot that could be operated at any scale - from the smallest pharmacy shop to the largest supermarket.

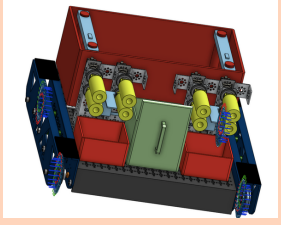
## STAGE 2: EXECUTION AND CAD

For the first few days we chose to work on individual mechanisms. We then entered a rapid prototyping phase, combining the chassis with these mechanisms as well as improving our existing mechanisms.
*Note: Our versions are marked using the "versions" feature in our CAD file. We utilised the "Part Studio" to create our own parts. We also utilised parts from the public FIRST library.*

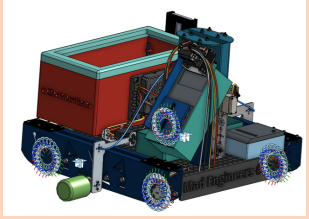
### VERSION 1



### VERSION 2



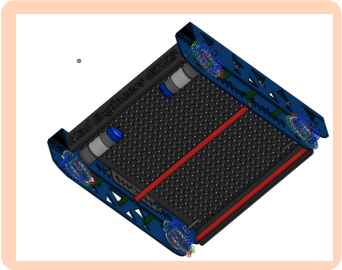
### VERSION 3



## STAGE 3: DEBUGGING AND OPTIMISATION

We visualised the **flow of the robot's movement** - from the moment the shopkeeper places the good till the robot collects the money and returns back. We **found a lot of little issues that we resolved** - the size of goods that could be placed, the strength of the supplies box, and the flow of the sanitiser from the reservoir. We also **later decided to include a billing machine in our design,** since it could be a potential vector for the virus. We even had to optimise for currency notes of different sizes, and figure out a unique mechanism to ensure that the same set of rollers could be used for cash as well as coin sanitisation.

# Robot Mechanisms

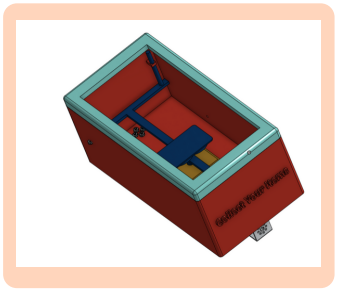


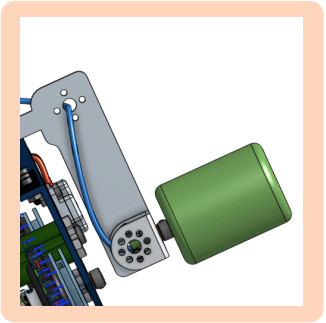
## CHASSIS DESIGN

We opted to **custom design** our chassis to give us a greater degree of freedom. With **triangular cuttings** to increase strength, the clean installation of the REV Expansion Hub, a **unique geared wheel design** that meshes with the pre-existing tracks, a powerful TETRIX® MAX TorqueNADO® Motor, ample beams, **churro tubings**, and plates to **ensure sturdiness**, our chassis serves our unique needs while remaining **robust and reliable**. The large plate covering the chassis gives us all the room we need to install our mechanisms.

## SUPPLIES BOX + PACKAGING

Our supplies box has a base size of **14" by 6**", giving ample room to store supplies. Our packaging mechanism detects whenever goods are placed in the box, due to the increase in tension in the spring. This triggers the release of an elastic bag (of any material), lowering the four attached pillars, which wraps the item. **The customer can then pick up this bag without having to touch the robot, the shopkeeper, or even the item itself.** This idea was actually inspired by a typical shoe cover dispenser machine.



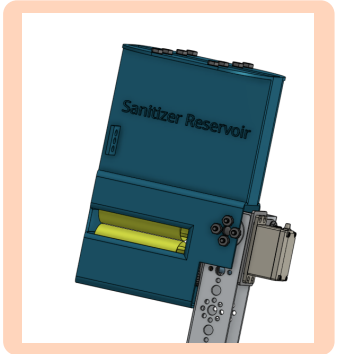


## TABLE SANITISATION

With a pair of retracting side rollers using a TETRIX® MAX Continuous Rotation Servo and a custom polycarbonate roller at the base of the robot, our robot is capable of **sanitising the entire table surface area.** The sanitising liquid is **obtained from a central sanitiser reservoir**. We utilise the natural force of gravity to deliver a pump-less design - the liquid flows from the higher reservoir to the lower rollers automatically. This mechanism kicks in after every single customer's transaction.

## CURRENCY SANITISATION

Cash and coin are inserted in the covered slots. There are **two porous sanitising rollers: both wet with sanitising liquid on the outer surface.** They are rotated by a Gobilda 2000 Series Dual Mode Servo (25-1). The cash and the coins can be collected after they pass through the polycarbonate roller manually. If they are not collected then, they are deposited in a tray on the sides of the sanitising mechanism.
The credit card and the billing machines are attached to the robot directly. A TETRIX® MAX Standard-Scale Servo Motor attached to a sponge regularly sanitises the number pad of the credit card machine.




| Strengths | Weaknesses |
|---|---|
| • ***Adaptability***: can be utilised with any countertop size in any shop with minimum configuration<br>• ***Ease of use***: Requires minimal user knowledge, intuitive mechanism placement means any one can use and understand<br>• ***Automation:*** Automatic packaging and movement means lesser manual labor from the side of shop-worker | • ***Turning***: Since the robot runs on a pre-defined track, it cannot turn in any other direction<br>• ***Tray size:*** The size of our supplies box is limited by FTC size guidelines. This may not be sufficient for some specific payloads. |